eNSP 中实现路由器与网络云（真实 PC 网卡）的桥接及 IPv6 通信

Implementing Bridging Between Routers and Network Cloud (Real PC Network Card) and IPv6 Communication in eNSP

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修订记录

日期	版本	描述	作者/修订人
2024-11-13	1.0	创建文档	朱仕耿

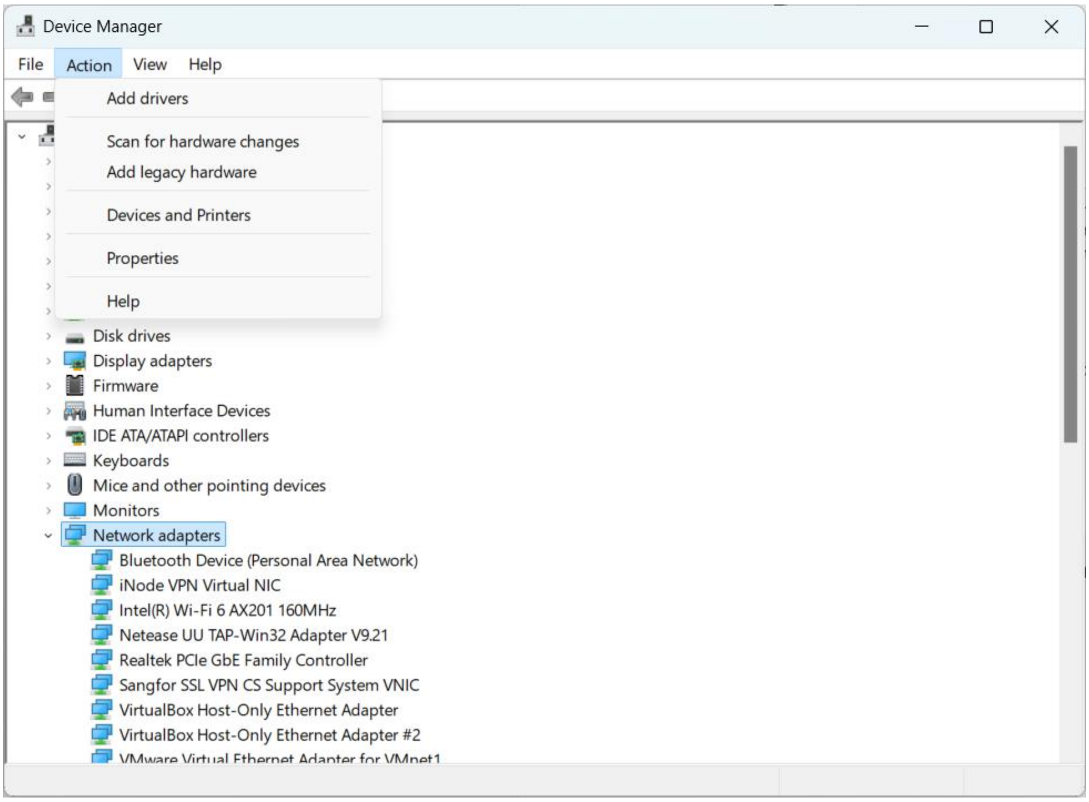
目 录

1 在 Windows 中添加环回网卡 Add a Loopback Network Adapter in Windows

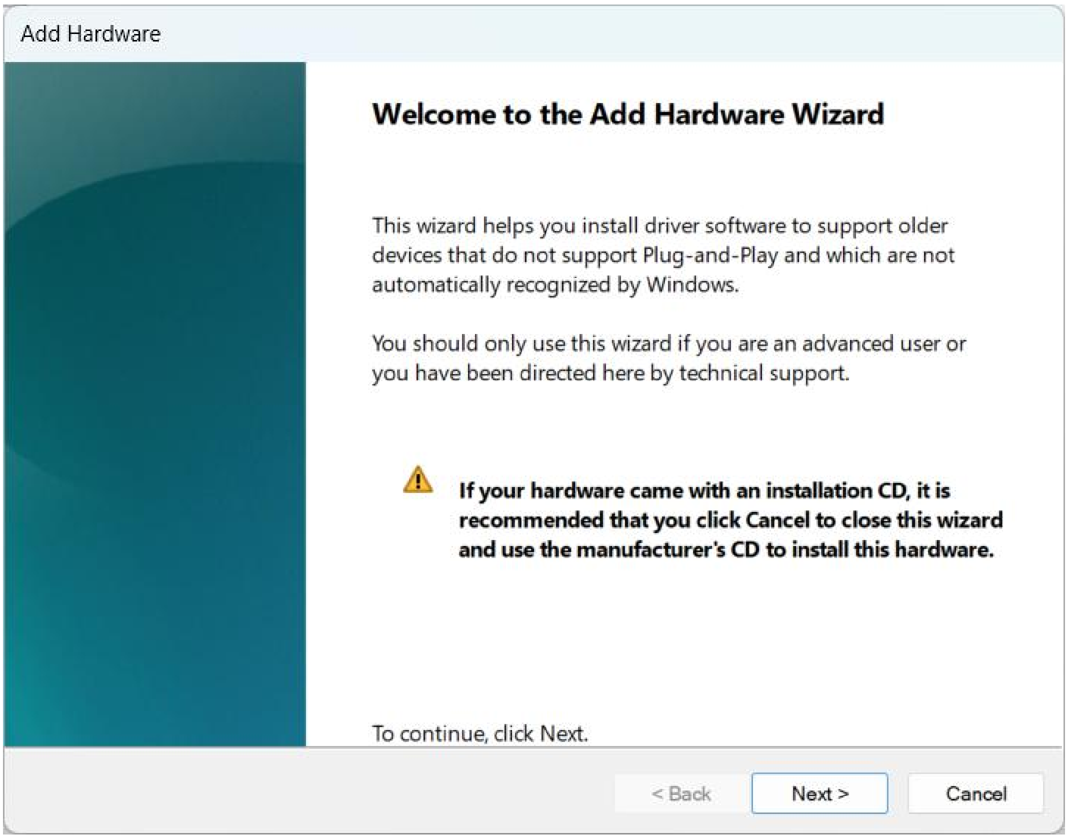
打开“设备管理器”。

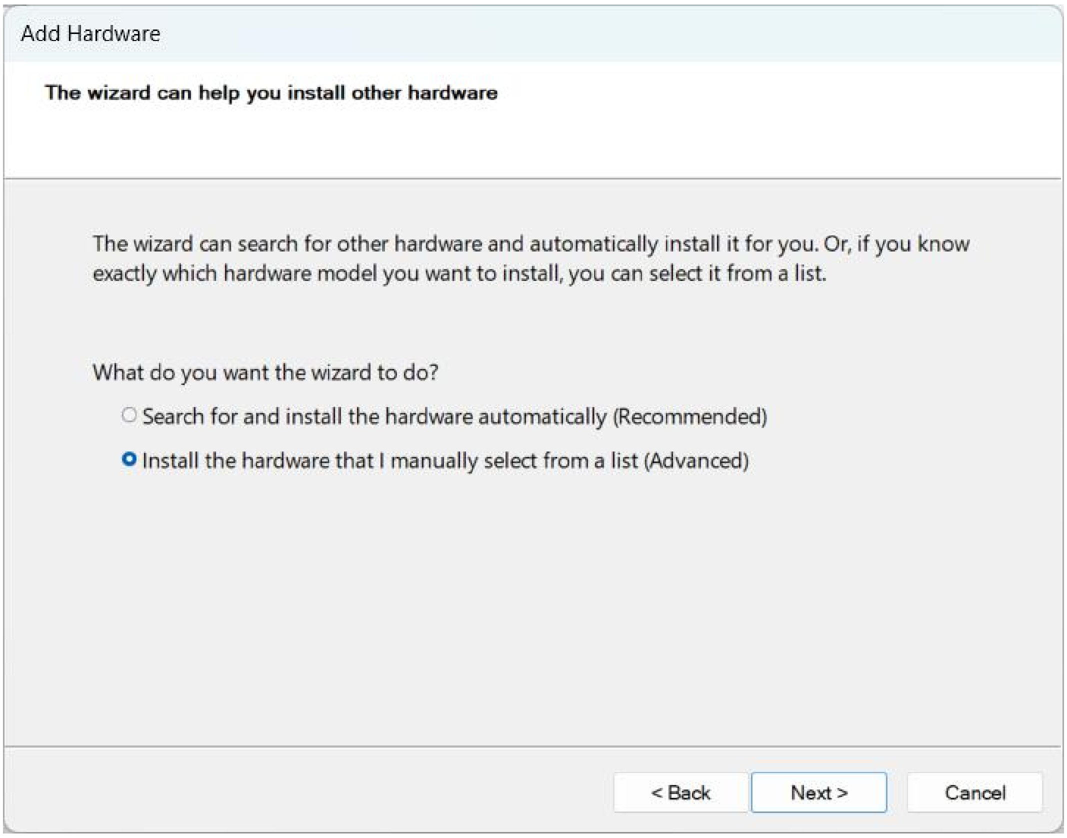
Open 'Device Manager'. (Win+x, then select ‘Device Manager’)

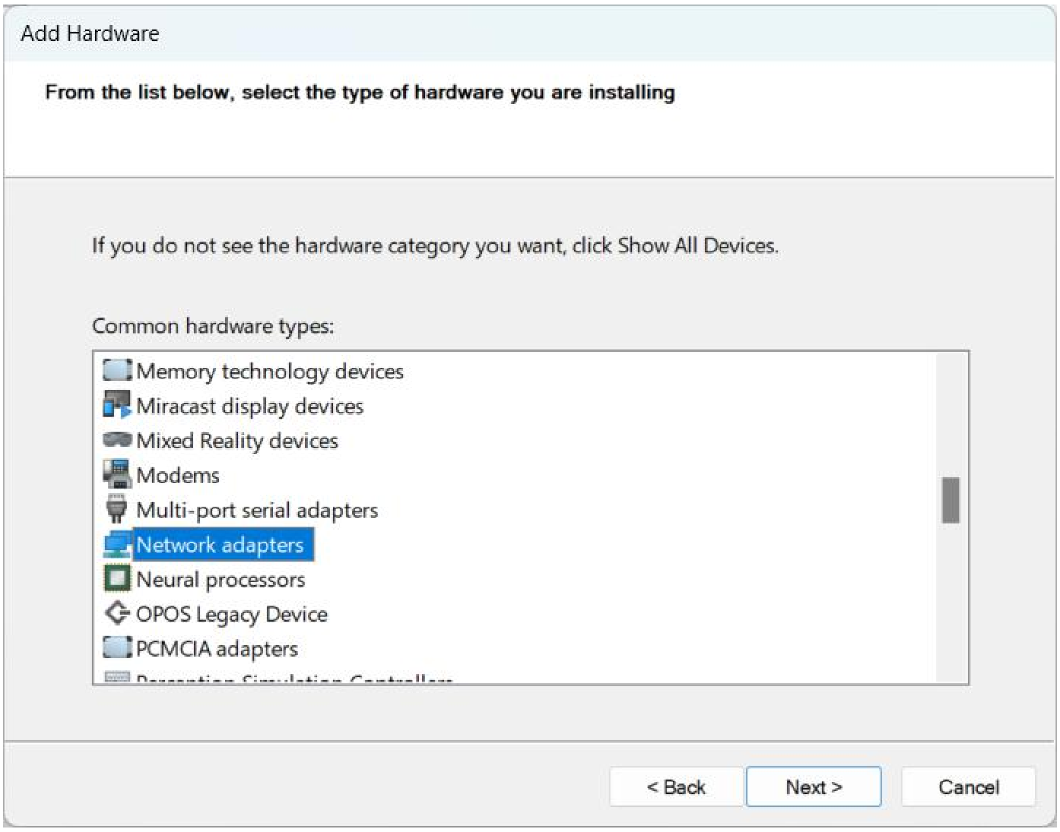
Select 'Network Adapters,' then click 'Action' > 'Add Legacy Hardware'.

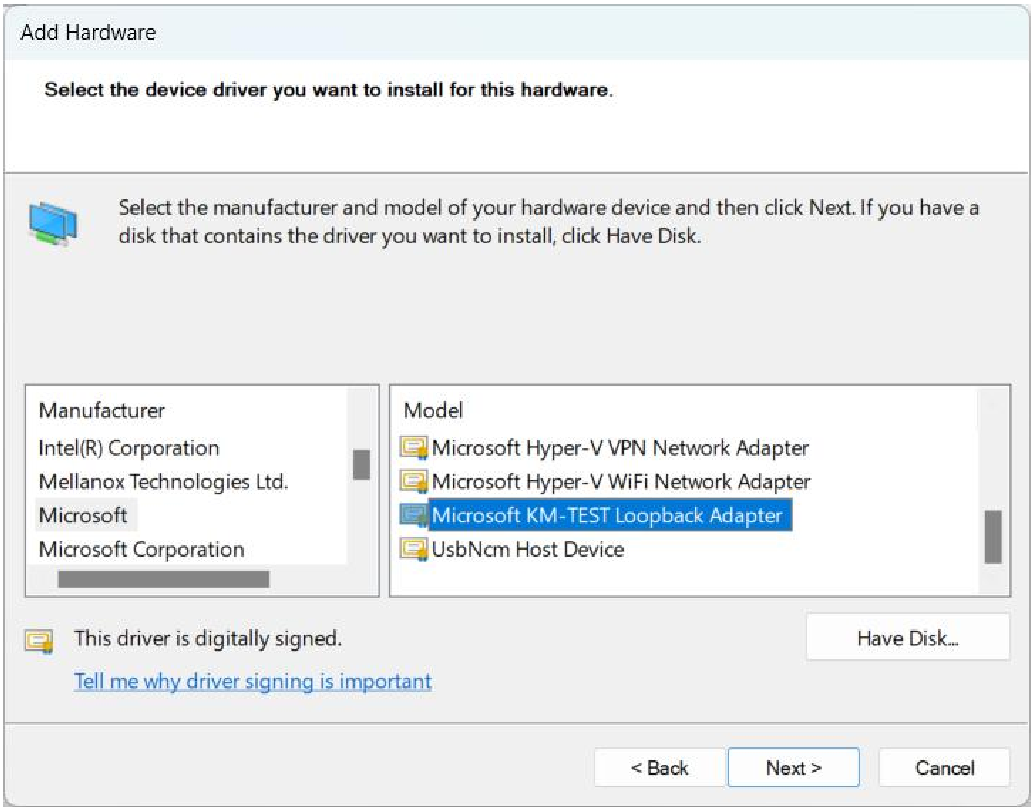


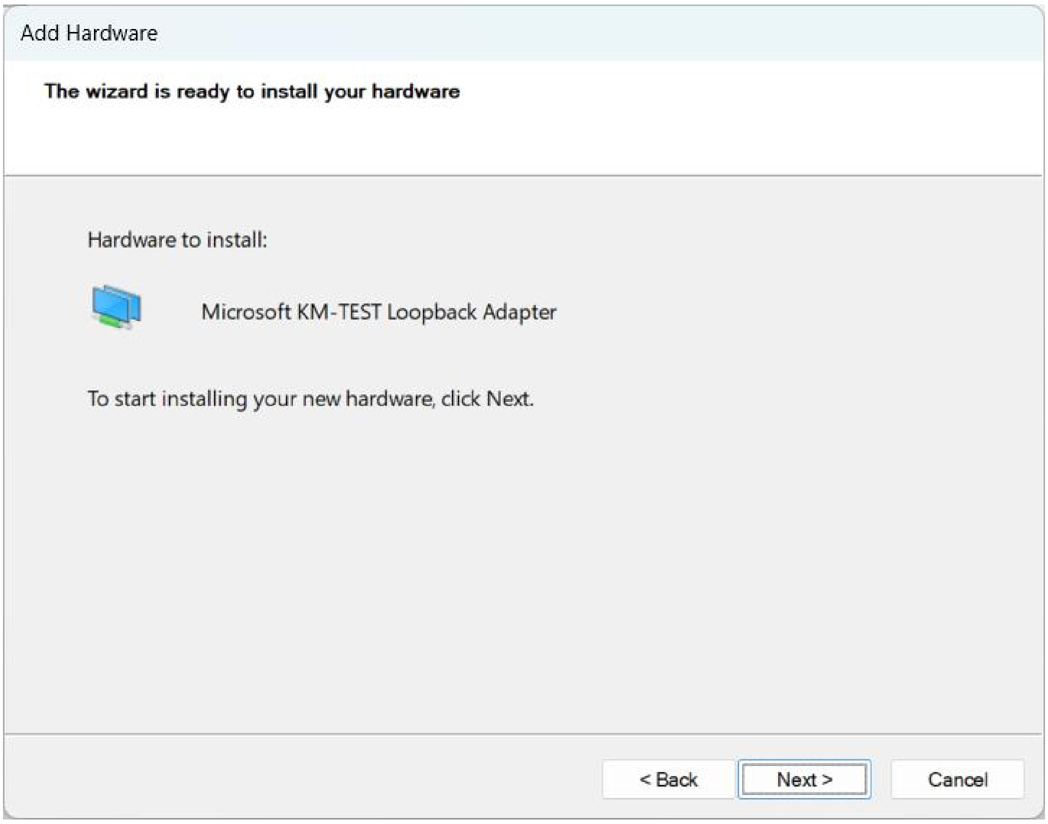
Add a loopback network adapter as shown in the figure below.




Add Hardware
The wizard can help you install other hardware
The wizard can search for other hardware and automatically install it for you. Or, if you know exactly which hardware model you want to install, you can select it from a list.
What do you want the wizard to do?
Search for and install the hardware automatically (Recommended)
Install the hardware that I manually select from a list (Advanced)
< Back
Next >
Cancel


Add Hardware
From the list below, select the type of hardware you are installing
If you do not see the hardware category you want, click Show All Devices.
Common hardware types:
Memory technology devices
Miracast display devices
Mixed Reality devices
Modems
Multi-port serial adapters
Network adapters
Neural processors
OPOS Legacy Device
PCMCIA adapters
< Back
Next >
Cancel

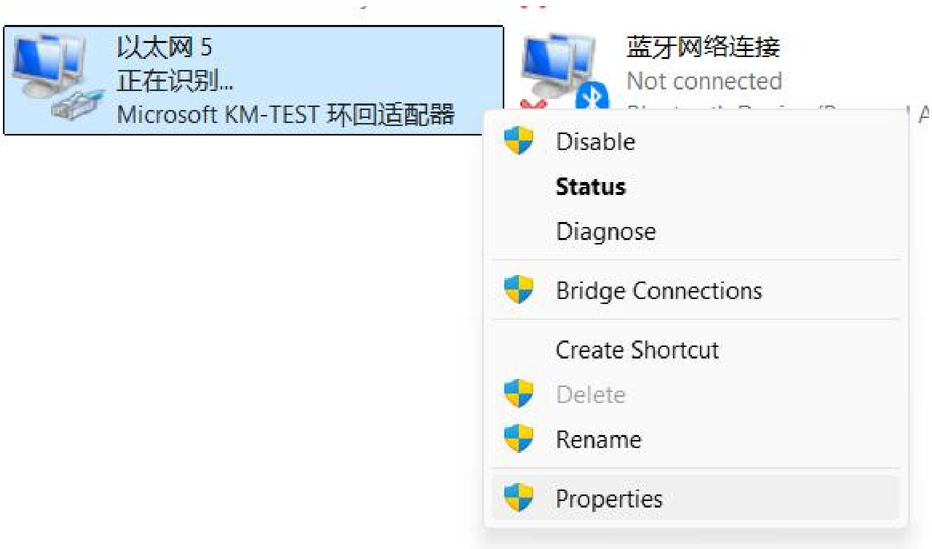


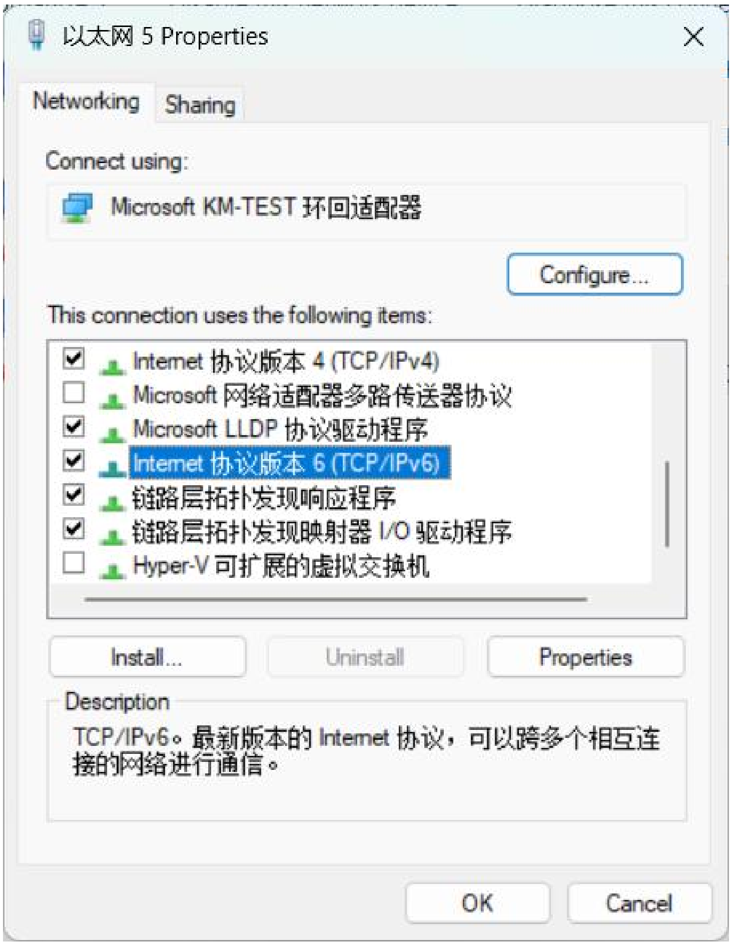


完成环回网卡添加后，为了让 eNSP 能够发现该网卡，需要重启电脑。

After completing the addition of the loopback network adapter, restart the computer to allow eNSP to detect the adapter.

2 配置环回网卡 IPv6 地址 Configure the IPv6 Address for the Loopback Network Adapter





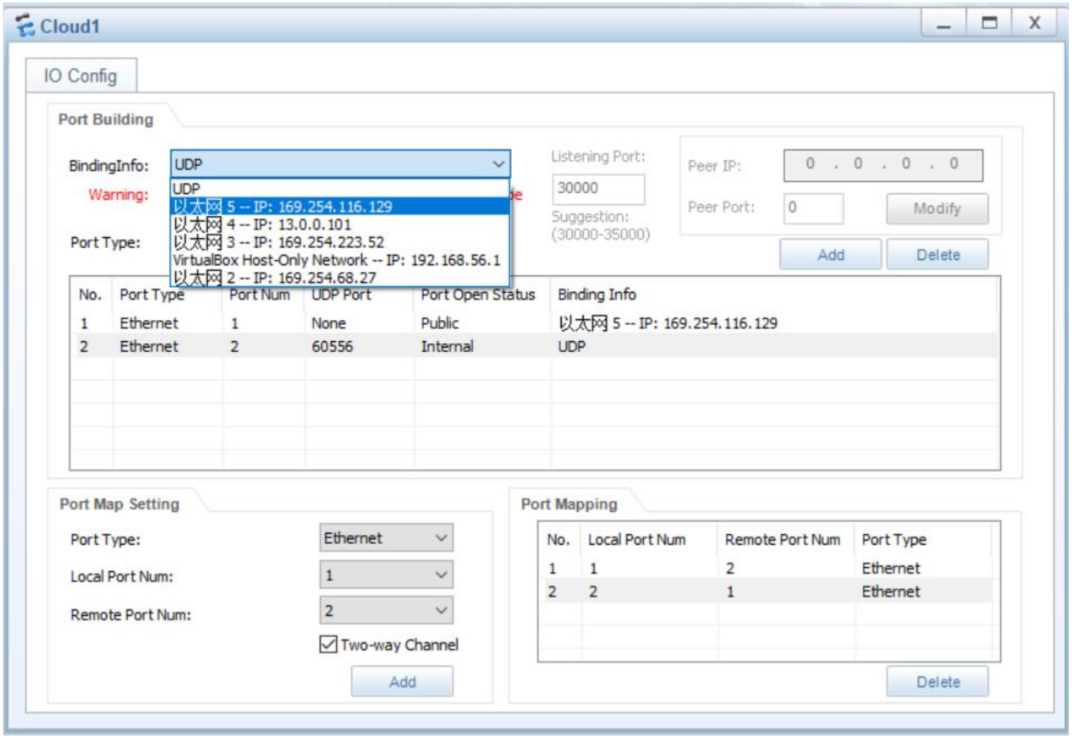
为网卡配置 IPv6 地址，略。

Configure the IPv6 address for the network adapter (details omitted).

3 在 eNSP 中创建拓扑
Create a Topology in eNSP

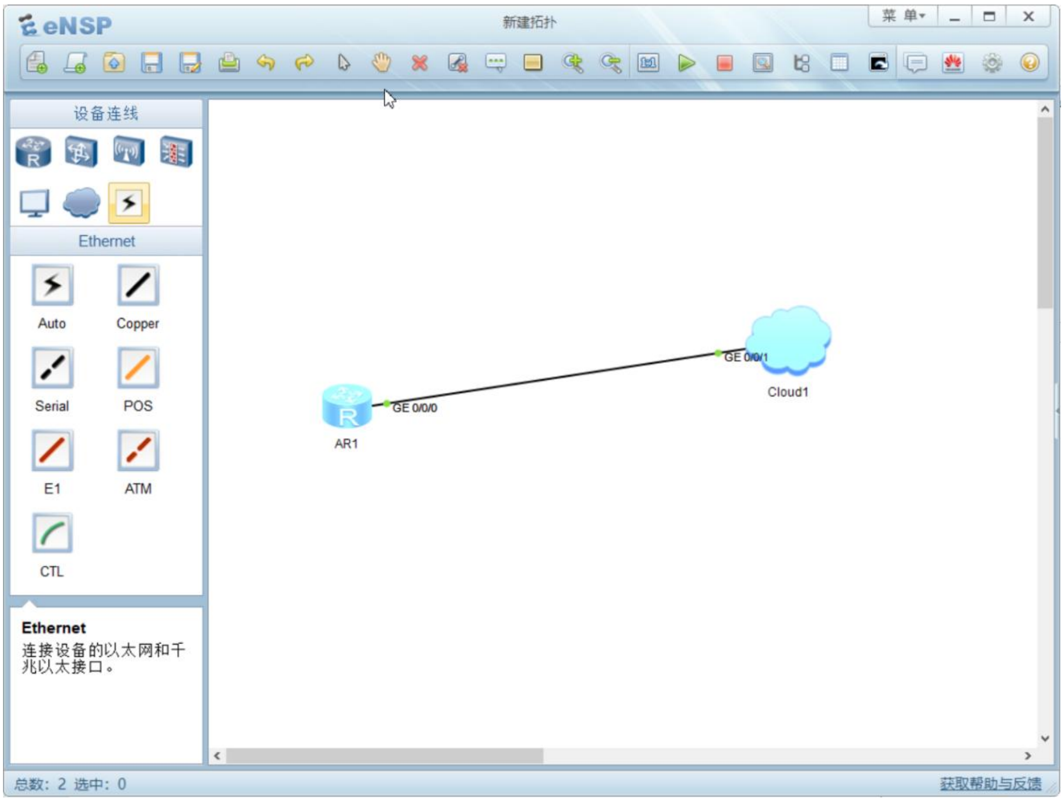
在拓扑中添加网络云，并按图示配置：

Add a cloud to the topology and configure it as shown in the diagram:



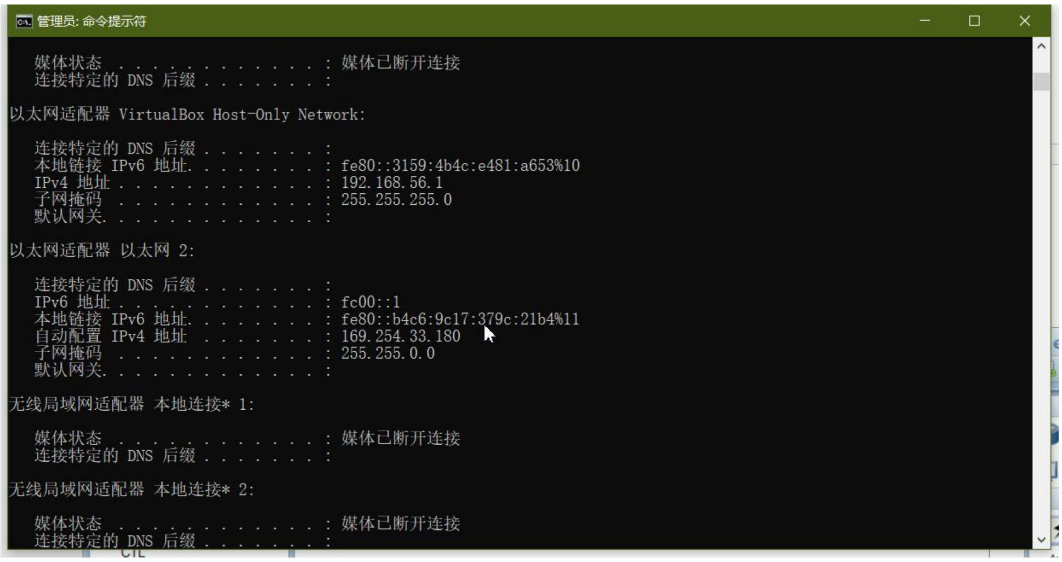
搭建拓扑。

Build the topology.



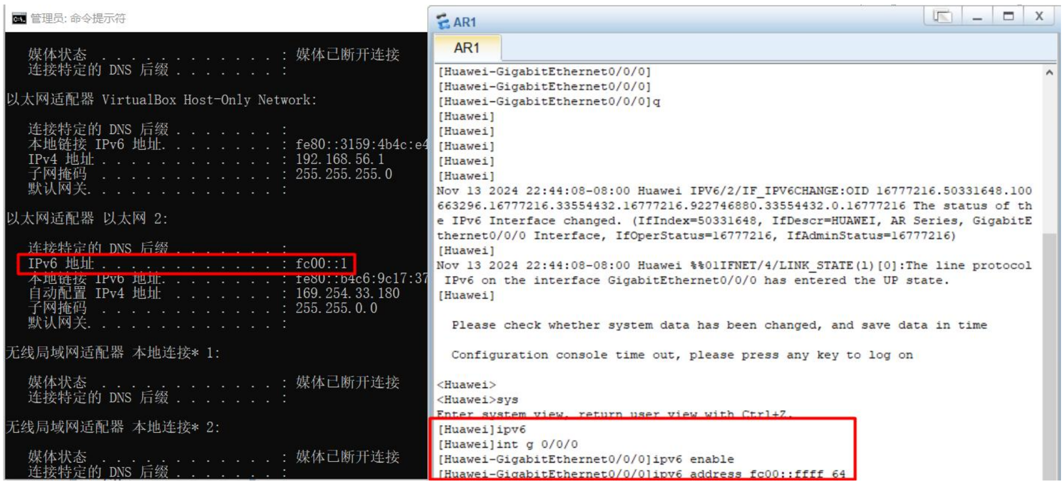
确认 PC 的环回网卡 IPv6 地址。

Verify the IPv6 address of the PC's loopback network adapter.



完成路由器 IPv6 配置。

Complete the IPv6 configuration on the router.


管理员: 命令提示符
媒体状态 . . . . : 媒体已断开连接
连接特定的 DNS 后缀 . . . . :
以太网适配器 VirtualBox Host-Only Network:
连接特定的 DNS 后缀 . . . . :
本地链接 IPv6 地址. . . . . : fe80::3159:4b4c:e4
IPv4 地址 . . . . : 192.168.56.1
子网掩码 . . . . : 255.255.255.0
默认网关. . . . . :
以太网适配器 以太网 2:
IPv6 地址 . . . . : fc00::1
自动配置 IPv4 地址 . . . . : 169.254.33.180
子网掩码 . . . . : 255.255.0.0
默认网关. . . . . :
无线局域网适配器 本地连接* 1:
媒体状态 . . . . : 媒体已断开连接
连接特定的 DNS 后缀 . . . . :
无线局域网适配器 本地连接* 2:
媒体状态 . . . . : 媒体已断开连接
AR1
AR1
[Huawei-GigabitEthernet0/0/0]
[Huawei-GigabitEthernet0/0/0]
[Huawei-GigabitEthernet0/0/0]q
[Huawei]
[Huawei]
[Huawei]
[Huawei]
[Huawei]
Nov 13 2024 22:44:08-08:00 Huawei IPV6/2/IF_IPV6CHANGE:OID 16777216.50331648.100
663296.16777216.33554432.16777216.922746880.33554432.0.16777216 The status of th
e IPv6 Interface changed. (IfIndex=50331648, IfDescr=HUAWEI, AR Series, GigabitE
thernet0/0/0 Interface, IfOperStatus=16777216, IfAdminStatus=16777216)
[Huawei]
Nov 13 2024 22:44:08-08:00 Huawei %%01IFNET/4/LINK_STATE(l)[0]:The line protocol
IPv6 on the interface GigabitEthernet0/0/0 has entered the UP state.
[Huawei]
Please check whether system data has been changed, and save data in time
Configuration console time out, please press any key to log on
<Huawei>
<Huawei>sys
[Huawei]ipv6
[Huawei]int g 0/0/0
[Huawei-GigabitEthernet0/0/0]ipv6 enable